# WEEK 5 OVERVIEW

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CEVE 543

2026-09-21

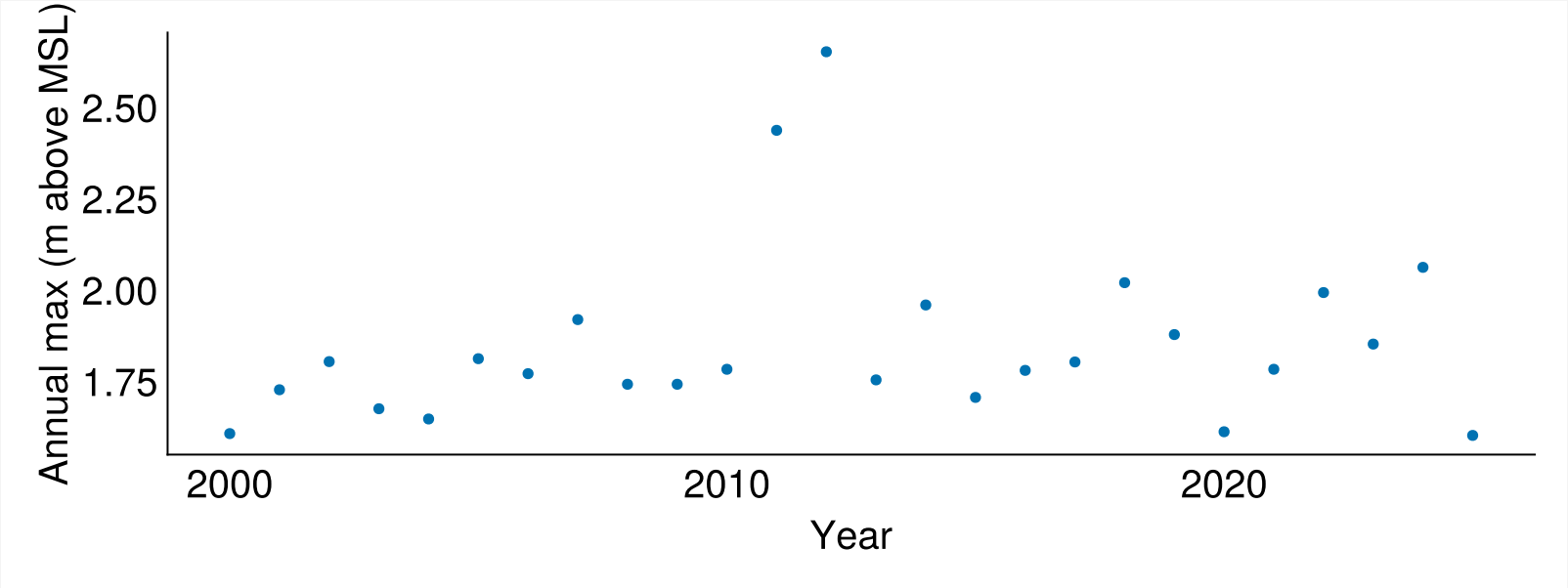


# THIS WEEK

- **Monday:** peaks over threshold and the GPD (today)
- **Wednesday:** practice problems + Brays Bayou case study
- **Friday:** lab 5, remote (no in-person meeting)

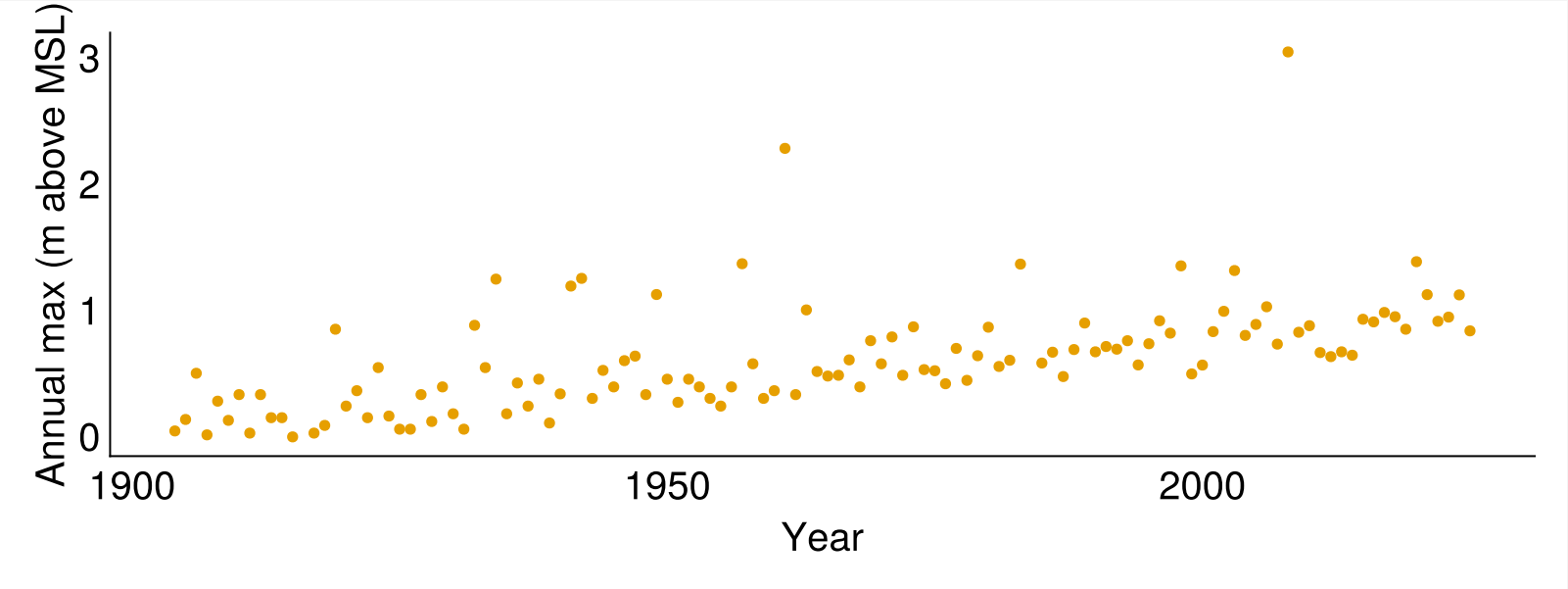
Lab 5 will be posted by Wednesday.

# THE PROBLEM WITH BLOCK MAXIMA




New Haven, CT (NOAA 8465705). Sandy (2012) and Irene (2011) tower over everything else. Do we have 26 annual maxima, or 2 real extremes and 24 ordinary high tides?

# SAME STORY, DIFFERENT COAST




Galveston Pier 21 (NOAA 8771450), 120 years of hourly data. Ike (2008, 3.0 m) and Carla (1961, 2.3 m) stand apart; everything else is under 1.4 m. Block

# SAME STORY, DIFFERENT COAST

maxima treats the year with Ike and a quiet year the same way: one observation each.

# WHY PEAKS OVER THRESHOLD?

Block maxima is **statistically wasteful** (Coles, §4.1). It reduces a year of sub-annual data to one number.

The HEC fishing analogy: block maxima keeps only the biggest fish each day. Peaks over threshold keeps every fish above the legal length.

Peaks over threshold uses **more of the data**, which means tighter confidence intervals on the same return level. The threshold is a modeling choice you can **diagnose** (mean residual life plot, parameter stability). “One year” is a convention nobody checks.

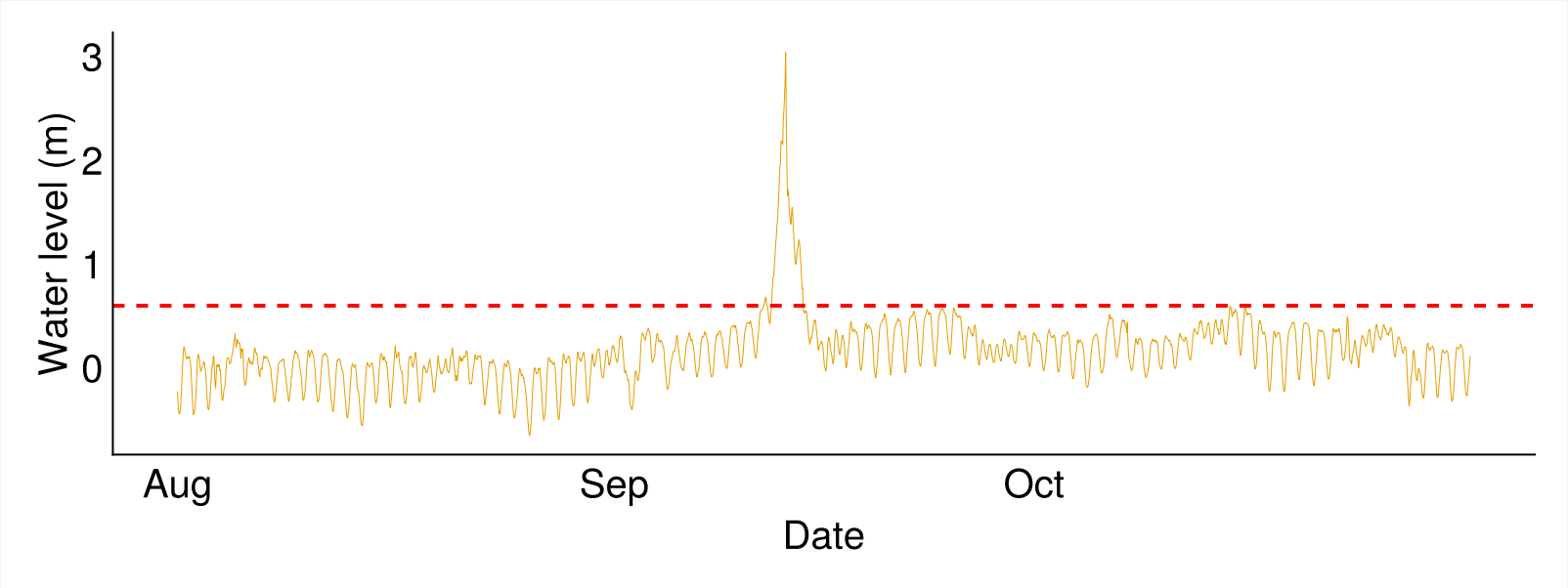
We need two ingredients:

1. **The distribution of exceedances** above the threshold (the GPD)

# WHY PEAKS OVER THRESHOLD?

2. **How often exceedances happen** (a rate, events per year)

# WHAT THE HOURLY RECORD LOOKS LIKE




Galveston, Aug–Oct 2008. The dashed line is a candidate threshold. Ike keeps the gauge above it for days: those are not independent exceedances.

# TODAY'S PLAN

1. The GPD and compound-Poisson derivation (board)
2. Threshold diagnostics worked on the Coles rainfall and Galveston (slides)

The connection between the GEV and the GPD, the compound-Poisson structure, and the judgment calls in choosing a threshold.